

plan

Implementation Plan: Authored AI/IT Knowledge Areas with Lessons, Tests and Video Anchors

Branch: main (this repository works on main only; 004-authored-ai-areas is the spec directory, not a branch) **Date:** 2026-09-07 **Spec:** [spec.md](#) **Input:** Feature specification from specs/004-authored-ai-areas/spec.md

Summary

Replace a catalogue of mined vocabulary with a catalogue of **authored AI/IT subjects**, each carrying lessons, supporting materials, video anchors into the recorded sessions, and a gated end-of-area test — and make every claim about that catalogue provable by a three-valued gate with a data-driven mutation proof.

A large part of this plan is already built and measured. This is not an estimate: the work landed from three parallel streams during the session that produced this specification. The plan therefore has an unusual shape — it records **what is done and how it was verified**, **what is done but unverified against real content**, and **what is not done** — because a plan that re-proposes finished work would hide the two gaps that actually matter.

The technical approach in one line: **one decision function, one authored content format, one time-link contract, and a gate for each** — with the publication decision computed once and consumed by every surface.

Technical Context

Language/Version: Go 1.26.2 (backend, workshop/platform/backend), TypeScript 5.x / Angular (frontend, workshop/platform/frontend), Python 3 (curriculum pipeline, workshop/pipeline/extract/), Bash (gates, workshop/platform/gates/) **Primary Dependencies:** submodules/curriculum-kit (the reusable Catalog → Area → Lesson → Material + Assessment library), the existing passage registry (workshop/curriculum/passages.jsonl), the existing transcript segment sidecars (start/end times), the design token source under design-system/ **Storage:** Files. Authored content as JSON under workshop/curriculum/learning/; taxonomy, materials, reviews and time links as JSONL/JSON siblings; learner progress as a session-keyed JSON store under the index volume **Testing:** Go go test (unit + handler gates), Playwright (frontend e2e), Bash three-valued gates with paired prove-*.sh mutation provers, Python unittest for the pipeline **Target Platform:** Linux container (podman), served on 127.0.0.1:8087; frontend bundle staged into platform/web/ **Project Type:** Web service + single-page application + content pipeline, inside a private module that must also clone standalone **Performance Goals:** A learner reaches a scored test result without any step reporting not-found; video anchors position within 1 s of the referenced moment (SC-008) **Constraints:** Content is PRIVATE and this umbrella is PUBLIC — no content may cross; the reusable library must stay project-unaware; every gate is three-valued and a 2 is never a pass; no CI, enforcement is local

NEEDS CLARIFICATION: none. The five open decisions were closed in [spec.md § Clarifications](#), each against a measurement. Two are flagged there as most likely to want operator override (the area-set boundary and the 70% default pass threshold); neither blocks this plan, because the first changes *which* areas are in scope and the second changes one declared number.

Constitution Check

GATE: Must pass before proceeding. Re-checked after design — see the second table at the end of this document.

Principle	Status	Notes
Evidence-Based Claims	PASS	Every figure in this plan is a captured run or a file measurement. The

Principle	Status	Notes
		three items that are not verified are named in <i>Known gaps</i> , not omitted.
		All four new gates are three-valued and each ships a prove-*.sh whose mutations are DATA read from an env var, never code edits.
Honest Instruments	PASS	Each prover carries a control mutation that must go green, and each carries a vacuity refusal that exits 2 — asserted by the prover itself.
		No universal clause is weakened. New gates register in platform/gates/check-
Governance Fidelity	PASS	registry-002.tsv; verify-check-registry-002.sh currently exits 0 at checks=94 debt=3.
		curriculum-kit is consumed by a replace directive pointing outside the repository, because it is a plain directory rather than a git repository and has no commit to pin.
Isolation by Default	NEEDS ATTENTION	Inside the umbrella this builds; a standalone clone does not. See <i>Known gaps</i> — the remedy is

Principle	Status	Notes
Comprehensive Documentation	PASS	upstream and is an operator decision. The time-link URL contract is written down (platform/frontend/docs/time-links.md) and is the producer's authority, not a convention.
Environment Adaptability	PASS	No absolute path is introduced. Gates build their own server from source on a kernel-assigned free port.
Authored Curriculum	PASS	The published set is the 37 authored subjects. An evidence-frequency floor was measured and rejected as a selector, because it made topicality worse (0 of 36 at floor 2 vs 32 of 814 at floor 1).
Published Means Served	NEEDS ATTENTION	Now enforced in code by a single <code>AreaPublicationOf</code>, and asserted end to end. But the running container still serves the old binary and the old 819; the served state is undetermined until it reloads. That is the principle's own failure mode and it is currently live.
	NEEDS ATTENTION	

Principle	Status	Notes
Derived Presentation		Titles, tags and summaries are derived rather than stored — which is exactly why null-titled mined areas rendered as single synthesised words. The derivation is correct; whether the served stylesheet reflects the widened palette is unverified.
Standalone Cloneable	NEEDS ATTENTION	verify-standalone-clone.sh went 2 gaps → 3, and the new row is the curriculum-kit replace target. This is the gate working, not drift.
Quality Over Speed	PASS	Three areas fail publication review on 62, 54 and 58 uncited claim blocks and are left failing . Stamping them would be the bypass.

No principle is violated. Four need attention, and all four are recorded in *Known gaps* with the measurement behind them. None is silenced.

Project Structure

Documentation (this feature)

```

specs/004-authored-ai-areas/
├─ spec.md           # Feature specification (393
lines)
├─ plan.md           # This file
├─ research.md       # Phase 0 – decisions, each with
the measurement that decided it

```

─ data-model.md	# Phase 1 – entities, invariants,
state	
─ contracts/	
─ catalogue.md	# Area list + detail, publication
reasons	
─ learning.md	# Lessons, materials, assessment,
submission	
─ time-links.md	# Video anchor URL grammar
─ authored-document.md	# The on-disk authored content
format	
─ quickstart.md	# Runnable end-to-end validation
─ tasks.md	# /speckit-tasks output
─ checklists/	
─ requirements.md	# Spec quality checklist (16/16)

Source Code (repository root)

submodules/curriculum-kit/	# The reusable library –
project-UNAWARE	
─ catalog.go area.go lesson.go	# Catalog → Area → Lesson
→ Material	
─ material.go anchor.go	#
VideoAnchor{chapterId,start,end,transcriptAnchor}	
─ assessment.go question.go	# Answers by ID, never by
index	
─ availability.go	# AvailabilityOf – the
single gate	
─ submit.go	# Submit; returns no
populated result when locked	
─ document.go	# DecodeDocument –
REJECTS unknown fields	
workshop/	
─ curriculum/	
─ learning/NN-<slug>.json	# ← AUTHORED CONTENT (the
remaining gap)	
─ taxonomy.jsonl	# extracted areas + terms
─ unpublished-areas.jsonl	# withheld rows, never
deleted	
─ area-materials.jsonl	# materials per area
─ publication-reviews.jsonl	# review per area
─ area-timelinks.json	# measured spans → video
anchors	
─ questions/*.senior.json	# authored senior
question banks	
─ docs/training/curriculum-areas/	# 37 authored area
documents	
─ pipeline/extract/	
─ measure_module_evidence.py	# evidence per module,
all chapters	
─ build_area_timelinks.py	# spans from segment

```

sidecars
|   └─ promote_curriculum_areas.py      # publication, via the
real minting bridge
|   └─ publication_policy.py             # the title-non-empty
rule
|   └─ verify_curriculum_areas.py       # gate, 11 data mutations
└─ platform/
    └─ backend/internal/api/
        └─ area_publication.go          # AreaPublicationOf – THE
decision, once
    └─ lessons.go                      # the six learning routes
    └─ backend/pkg/learning/           # authoring layout,
loader, session store
    └─ frontend/src/app/               # area detail, lessons,
test, transcript
    └─ frontend/docs/time-links.md     # the URL contract
    └─ gates/verify-*.sh prove-*.sh    # 23 gates / 27 proofs
and growing
    └─ qa/fixtures/learning/           # synthetic corpus –
every byte invented

```

Structure Decision: The reusable shape lives in curriculum-kit and knows nothing about this project; pkg/learning is the only consumer that knows both. That boundary is enforced by construction rather than by discipline — curriculum-kit has an empty require set, so a project import would not compile. pkg/learning contains **no second availability test, scoring routine or answer comparison**; AvailabilityOf decides availability and Submit (never the ungated Grade) decides whether a submission is allowed and what it scores.

Execution Strategy

TDD Requirements



The publication decision — TDD, and it is the reason D3 existed. Both handlers had passing unit tests; nothing tested the *pair*. TestAreaListAndDetailAgree is the test that had to exist. **Done.**



The assessment gate — TDD. The load-bearing mutation is B4/M7, “the gate opens as soon as *any* lesson completes”: it passes B1, B2, B3, B5 and B7 and is caught **only** by B4, which is why the fixture’s required set must be a proper subset of its lessons. **Done, 17 mutations.**



Answer-key non-disclosure — TDD, with a leak check that fails if correct_choices or explanation appears on the taking path. **Done.**



The authored-document loader against real content — the loader rejects unknown fields, so the first authored file either decodes or fails loudly. Write one file, prove it loads, then write the rest. **In flight.**



Contrast and hue measurement of the SERVED bundle — the check must read what is served, not what is in the token source. **Not done.**

Parallel Execution Opportunities

Four streams have run concurrently and their file sets were disjoint by construction. They are recorded because the partition is reusable:



Content authoring (docs/training/, curriculum/, pipeline/extract/) — no platform/** writes.



Backend serving (platform/backend/**, platform/gates/**) — no curriculum/** writes.



Frontend (platform/frontend/src/app/**) — components and routes.



Theming (platform/frontend/src/styles/**) — stylesheets only.

One collision was observed and is the lesson: the frontend stream measured a stylesheet the theming stream was mid-rewrite of, and read a transiently unbalanced file. The partition held; the *timing* did not. Any stream that measures another stream's output must rebuild immediately before measuring, and say which build it measured.

Human Checkpoints

1. **Before the container restarts** — the restart is what makes the on-disk work served, and it is also what invalidates every “before” measurement. Do it once, deliberately, after all writers have stopped.
2. **Before any area is marked published without a review** — three areas fail on uncited claim blocks. They stay failing until a human reads them.

3. **Before curriculum-kit is published upstream** — that closes the standalone gap and is an operator decision.
4. **Before the 70% pass threshold is fixed** — the only decided value in the spec with no measurement behind it.

Review Gates



The publication decision function — every surface consumes it; a second implementation is the defect class it exists to prevent.



The time-link URL grammar — a producer that emits a malformed span produces a link that looks fine and is broken.



The authored document format — reviewed before 37 files are written to it, not after.

Known gaps — what is NOT done, and what is done but unverified

Recorded here rather than in a footnote, because these are the difference between this plan and a completion claim.

Not done:

1. **No authored learning content exists.** Every real area answers a determined `no_learning_catalog`. The six learning routes are proved against a synthetic bicycle-maintenance fixture and **have never been exercised against authored curriculum content.** Dispatched; in flight.
2. **The served palette is unmeasured.** Token files widened the hue span; no check reads the served stylesheet. SC-010 and SC-011 have no instrument yet.
3. **difficulty and category on questions cannot be closed here** — no such field exists in curriculum-kit. Upstream change required. Not fabricated.
4. **Spaced repetition is not ported.** The kit declines it deliberately; the interviewing module writes `streak/due_at` and reads neither.
5. **Chapter 02 was never mined into this taxonomy.** A topic taught only there, in words the chapter-01 lexicon lacks, is invisible to every coverage figure.

Done but unverified, and therefore not a pass:

1. **The running container serves the old binary and the old 819 areas.** `verify-server-unity.sh` is RED against it at 6 FAIL — all six the new routes answering a plain-text 404. Against a server built from this tree the same gate is PASS=43 FAIL=0 UNDET=0 DEBT=8. This clears on restart.
2. **The 8 diagram SVGs were not rasterised.** Well-formedness, theming and absence of external references are verified; text fit at a given font metric is not.
3. **unpublished-areas.jsonl holds 1,128 rows, a superset of the 814 actually withheld.** The exact 814 cannot be identified — the pre-filter taxonomy is git-ignored and was not retained.

Blocked on an operator decision:

1. **submodules/curriculum-kit is not a git repository.** It has no commit, so no truthful `helix-deps.yaml` entry can be written and a fabricated `ref:` would be a bluff. Recorded verbatim in `go.mod`.

Post-design Constitution re-check

Re-evaluated after Phase 1 produced `data-model.md` and `contracts/`.

Principle	Before design	After design	What changed
Evidence-Based Claims	PASS	PASS	The contracts state which figures are measured and which are undetermined.
Honest Instruments	PASS	PASS	Unchanged.
Governance Fidelity	PASS	PASS	Unchanged.
Isolation by Default	NEEDS ATTENTION	NEEDS ATTENTION	Unchanged — the remedy is upstream.
Comprehensive Documentation	PASS	PASS	The authored-document format is now written down before 37 files

Principle	Before design	After design	What changed
			are written to it.
Environment Adaptability	PASS	PASS	Unchanged.
Authored Curriculum	PASS	PASS	Unchanged.
			The design adds a served corpus revision (FR-023a) so the discrepancy becomes detectable; it does not by itself resolve the live one.
Published Means Served	NEEDS ATTENTION	NEEDS ATTENTION	Unchanged until a served-bundle instrument exists.
Derived Presentation	NEEDS ATTENTION	NEEDS ATTENTION	Unchanged.
Standalone Cloneable	NEEDS ATTENTION	NEEDS ATTENTION	Unchanged.
Quality Over Speed	PASS	PASS	Unchanged.

No gate failed. No unresolved clarification remains. Ready for / speckit-tasks.